VIETNAM NATIONAL UNIVERSITY HO CHI MINH CITY
UNIVERSITY OF INFORMATION TECHNOLOGY



HOMEWORK - LAB 03

Drum Machine, Grand Piano 88 and PVS Arena

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Chapter 1: Homework 1: Drum Machine

7 keyboard drum pads (`e.code`, no key repeat), polyphonic playback, hit log, level meter, plus Controls (kit, volume, mute) and a Looper (record, overdub, loop/once, speed).

1.1 Evidence

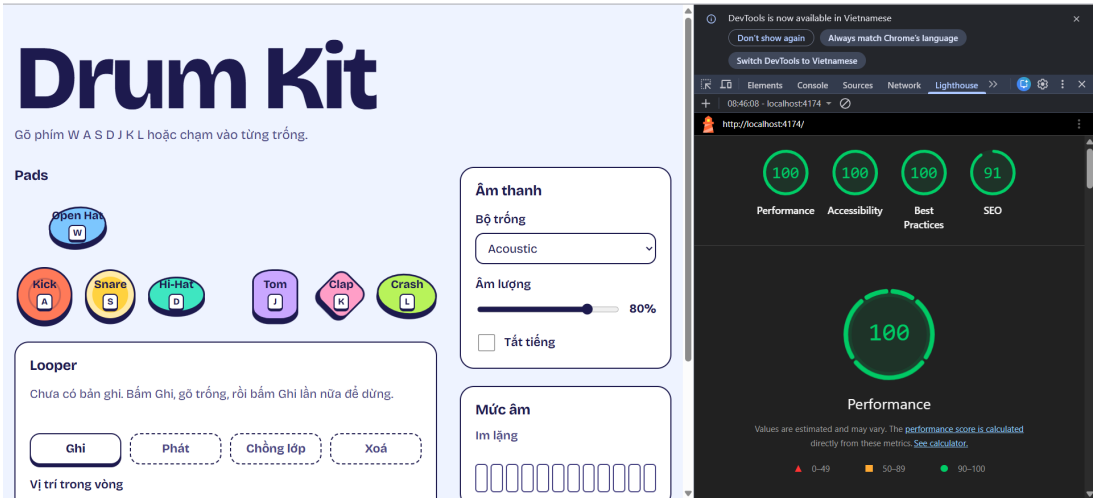


Figure 1.1: Lighthouse on the production build: Performance 100, Accessibility 100, Best Practices 100, SEO 91

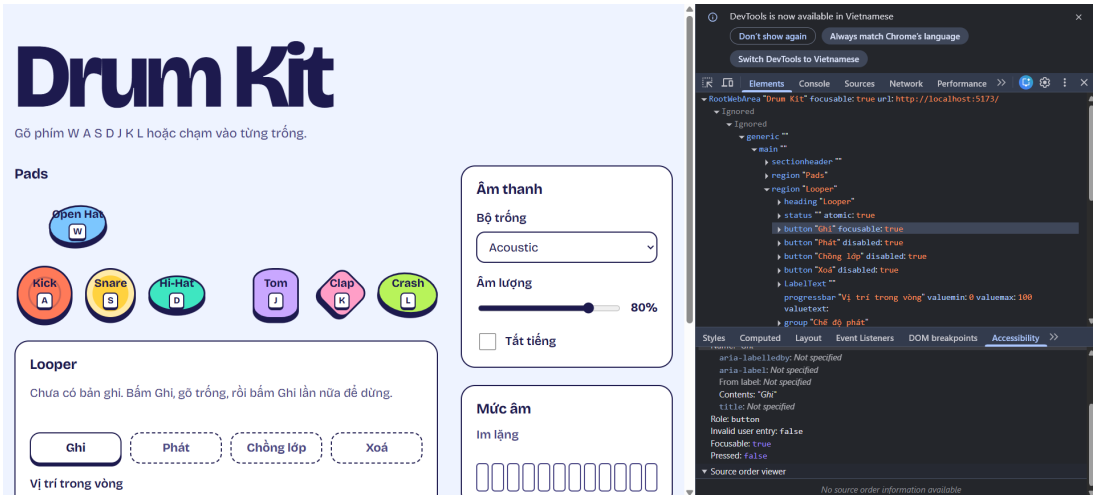


Figure 1.2: Accessibility tree of the Looper region: named buttons, `progressbar`, `status`

1.2 Verification

Check	Result
`npm test`	15 files, 146 tests pass
`npm run lint`	Clean
Contrast (WCAG 1.4.3, 1.4.11)	16/16 pairs pass, lowest 6.24:1
Lighthouse Performance	69 (dev server) $\rightarrow$ 100 (build)
Lighthouse Accessibility	100

Table 1.1: Homework 1 results

Not done: Lighthouse JSON export, 375 px and keyboard-tab screenshots, screen reader test. The Performance drop to 69 was caused by AI-added Google Fonts and eager WAV preloading (AI failure audit, error 5); fixed and re-measured.

1.3 Source Code

View Homework 1 on GitHub

Chapter 2: Homework 2: Grand Piano 88

88-key piano (52 white, 36 black), polyphonic synth with at most 8 voices and FIFO voice stealing, memoized keys, and a recorder with abortable replay at 0.5x, 1x and 2x.

2.1 Evidence

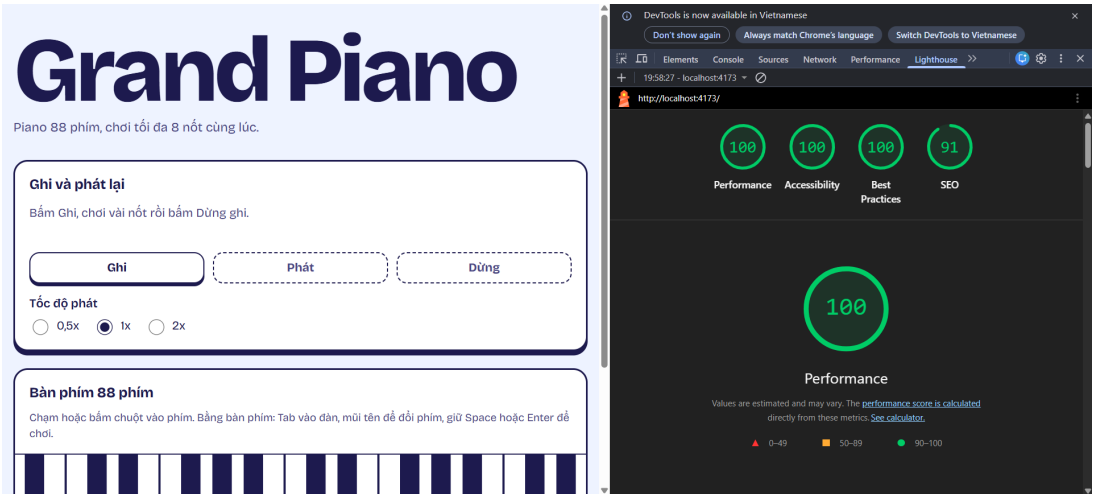


Figure 2.1: Lighthouse on the production build: Performance 100, Accessibility 100, Best Practices 100, SEO 91

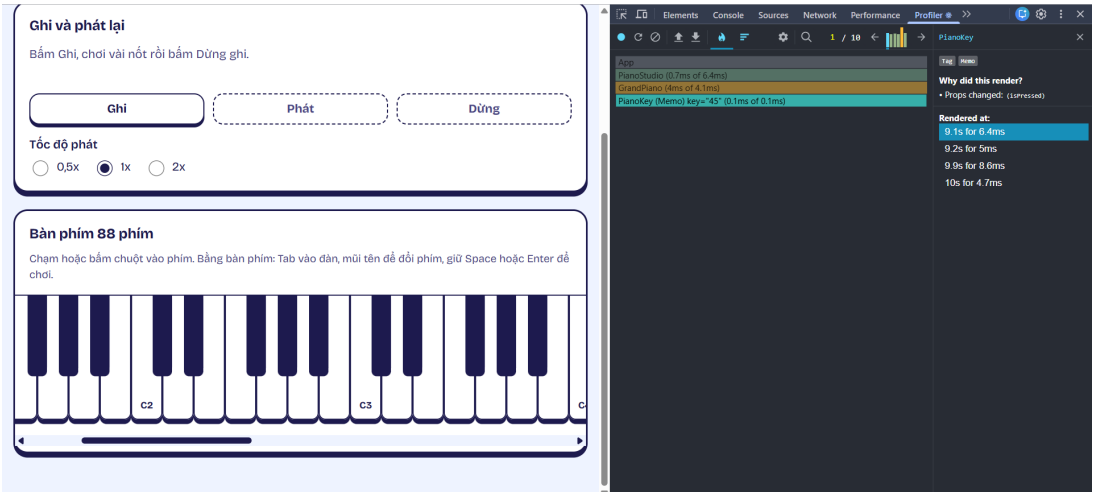


Figure 2.2: React Profiler: one key press re-renders one `PianoKey` (reason: `isPressed`)

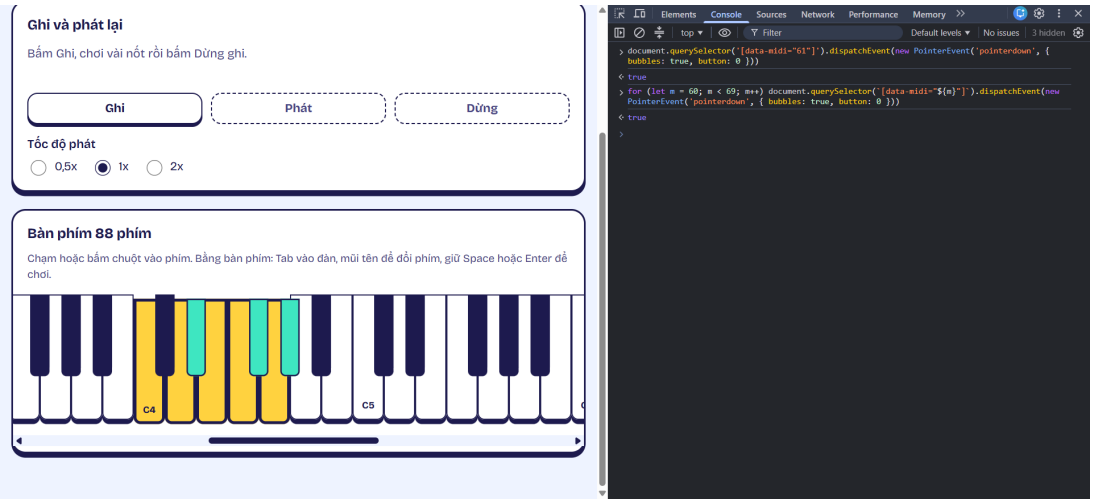


Figure 2.3: 9 keys pressed: 8 stay lit, the oldest key was stolen (FIFO)

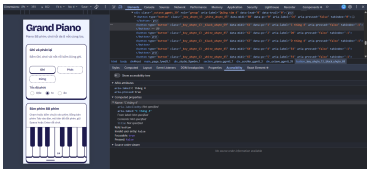


Figure 2.4: 393 px viewport: only the keyboard area scrolls

2.2 Verification

Check	Result
`npm test`	10 files, 140 tests pass
`npm run lint`	Clean
Frequency of A4, A0, C8	440 Hz, 27.5 Hz, 4186.01 Hz
Contrast	13/13 pairs pass, lowest 10.15:1
Lighthouse (build)	100 / 100 / 100
Profiler, one key press	One `PianoKey` re-rendered

Table 2.1: Homework 2 results

Not done: 375 px screenshot (the one taken is 393 px), keyboard-tab and reduced-motion screenshots, listening test, screen reader test. The Profiler screenshot predates wrapping `Sequencer` in `memo`; a test now covers it. WBS items H2-11 and H2-12 stay unticked for these reasons.

2.3 Source Code

View Homework 2 on GitHub

Chapter 3: Homework 3: PVS Arena

Airdrop crates with a drift-free countdown, optimistic claim with rollback on HTTP 409, double-click protection, VND prices and a sortable, filterable leaderboard.

3.1 Evidence

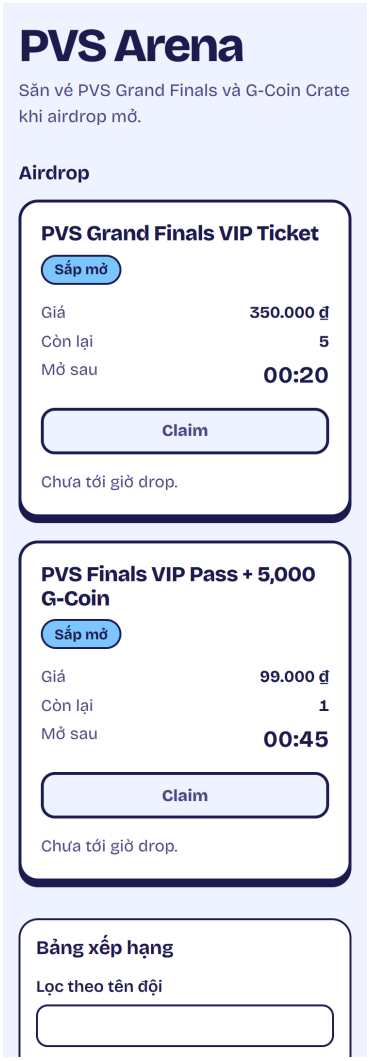


Figure 3.1: 375 px (top of the page): countdown, VND price, disabled Claim with a reason

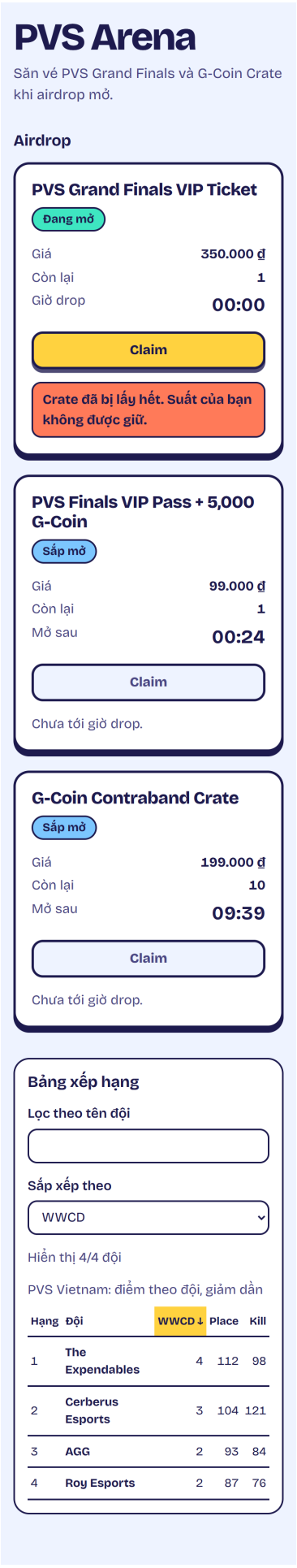


Figure 3.2: Claim rejected with 409: stock returns to its old value and an alert explains why

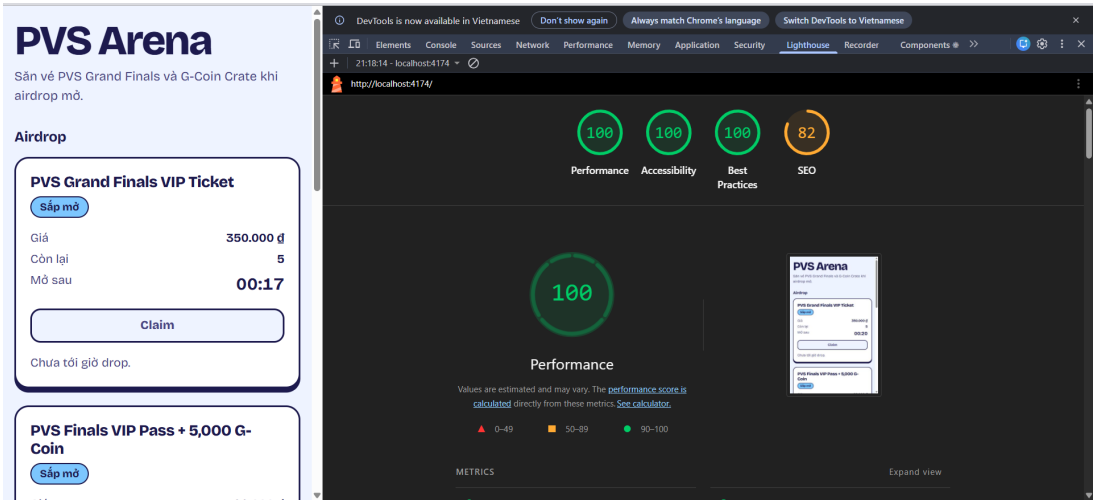


Figure 3.3: Lighthouse: Performance 100, Accessibility 100, Best Practices 100

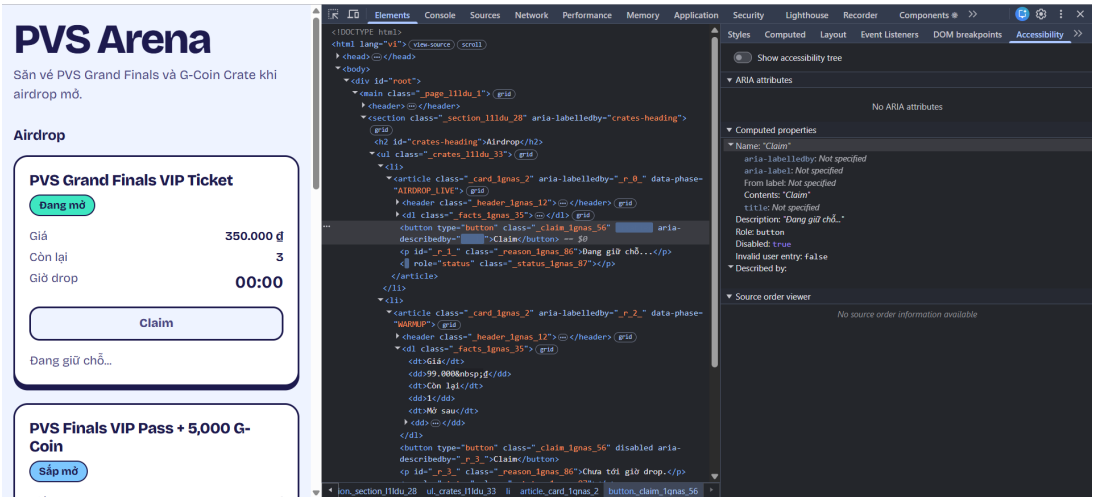


Figure 3.4: Accessibility tree of the Claim button

3.2 Verification

Check	Result
`npm test`	7 files, 61 tests pass
`npm run lint`	Clean
Hidden tab 330 s	Displayed 04:29, drift 0 s (before and after wake)
409 rollback	Stock 1 → 0 → 1, alert shown
Horizontal scroll, 320 and 375 px	None
Contrast	14/14 pairs pass, lowest 6.24:1
Claim transition	0.07 s; 0 s with reduced motion
Lighthouse	100 / 100 / 100

Table 3.1: Homework 3 results

Not done: screen reader test; Lighthouse SEO was not run.

3.3 AI Failure Audit

- Test runner reported as flaky"; the real cause was running `npm test` from the wrong folder (error 6).
- Countdown `<output>` was a live region announcing every second (error 7).

3.4 Source Code

View Homework 3 on GitHub

View AI failure audit on GitHub

Chapter 4: Submission & Resources

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- **Tests:** 347 pass in total (HW1 146, HW2 140, HW3 61); lint clean in all three.
- **Prompt logs:** `docs/prompt_logs/`
- **AI failure audit:** `AI_FAILURE_AUDIT.md`
- **Evidence:** `evidence/` in each homework folder
- **Limits:** no screen reader testing; HW1 and HW2 still lack some 375 px and keyboard screenshots.

END